Shuyue Jia (Bruce)

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Research Statement

I am an AI researcher with production engineering experience, focusing on both fundamental and applied AI. Please visit my Google Scholar (**with >950 citations, >10 H-index**) to explore our team's recent publications.

- **Text/Image/Video Processing**, **Understanding**, **Generation**, and **Quality Assessment**
- **Multimodal Foundation Model (FM)** and **Large Vision-language Model (LVLM)**
- **Large Language Model (LLM)**
- **Autonomous AI System**, **LLM/LVLM-based AI Agent**, and **Multi-agent System (MAS)**
- **AI for Human Health**
- **Retrieval-augmented Generation (RAG)** and **Cross-modal Retrieval**
- **Knowledge Graph** and **Information Retrieval**
- **EEG-based Brain-computer Interface (BCI)**

I am dedicated to advancing frontier and fundamental AI technologies that effectively and efficiently address real-world challenges. My research aims to translate cutting-edge research and scalable AI systems into real-world impact and clinical value. My ultimate goal is to **develop safe, reliable, and extensible Artificial General Intelligence (AGI) systems for Medicine** that **bring value to our society**. As part of my research, I actively contribute to software products, develop AI toolkits, and create benchmark datasets to drive progress in the field. All source codes from my work will be openly shared on GitHub (**with >2K stars**), trained models and datasets will be available on Hugging Face, and applied research in Vision and Language will be developed into **products**. Additionally, I have hands-on experience in **large-scale AI model training**, **large-scale AI model deployment**, and **large-scale AI-powered data processing**. **I am dedicated, hardworking, and always striving for excellence.**

Services and Awards

Chair, IEEE Boston Section Circuits and Systems Society, Boston, MA	2026 - Present
Program Committee Member, Association for Computational Linguistics (ACL) Industry Track, San Diego, CA	2026
Technical Committee Member, Embodied AI Systems, IEEE Systems, Man, and Cybernetics Society	2026 - Present
Track Chair, IEEE MIT Undergraduate Research Technology Conference (URTC), Cambridge, MA	2025 - Present
Speaker Program Chair, IEEE International Conference on AI and Data Analytics (ICAD), Boston, MA	2025 - Present
Scientific Program Committee Member, 2025 AMIA Clinical Informatics Conference, Anaheim, CA	2025
2025 IEEE Boston Section Arthur Winston Student Award, IEEE Boston Section	2025
Wiley Top Cited Article 2023-2024, Wiley Top Cited Papers, Wiley	2025
Local Conference Committee Member, IEEE Boston Section, Boston, MA	2024 - Present
Distinguished Computer Engineering Fellowship, Boston University	2023 - 2024
Research Assistantship, City University of Hong Kong	2021 - 2023
Innovation Scholarship, Northeast Electric Power University	2017 - 2019
Excellent Student Scholarship, Northeast Electric Power University	2016 - 2020
2019 Interdisciplinary Contest In Modeling, USA ***Honorable Mention***	Apr 2019
2018 Mathematical Contest In Modeling, Jilin, China ***First Prize***	Aug 2018
The 32^{nd} Chinese Physics Olympiad (CPhO), China ***Third Prize***	Oct 2015
2015 National High School Math League, China ***Second Prize***	Sept 2015

Academic Appointments

Boston University, MA, USA **Research Assistant** May 2024 - Present

- Supervisor: Prof. Vijaya B. Kolachalama
- Projects: Scientific & Medical Foundation Model, AI Grounding, Specialized Intelligence, and Personal Intelligence

City University of Hong Kong, Hong Kong, China **Research Assistant** Sept 2021 - Apr 2023

- Supervisor: Prof. Shiqi Wang
- Projects: Image Quality Assessment (IQA) and Perceptual Optimization

Education

Boston University, Boston, MA, USA Sept 2023 - May 2028 (expected)

- Ph.D. Candidate in Computer Engineering, Department of Electrical and Computer Engineering
- **Supervisor**: Prof. Vijaya B. Kolachalama
- **Expertise**: Scientific & Medical Multimodal Foundation Model and Medical Imaging with AI

City University of Hong Kong, Hong Kong, China May 2021 - June 2023

- M.Phil. in Computer Science, Department of Computer Science
- **Supervisor**: Prof. Shiqi Wang
- **Expertise**: Image Quality Assessment, Perceptual Optimization, and Deep Learning
- **Thesis**: No-reference Image Quality Assessment via Non-local Modeling

Northeast Electric Power University, Jilin, Jilin, China Sept 2016 - June 2020

- B.Eng. in Intelligence Science and Technology, School of Automation Engineering
- **Supervisor**: Prof. Yimin Hou and Prof. Jinglei Lv
- **Thesis**: Brain-computer Interface Signals Classification and Its Applications based on Deep Learning Methods

University of California, Irvine, Irvine, CA, USA Jul - Sept 2017

- Visiting Student, Department of Computer Science
- Selected coursework: Computer Systems and Architecture (A+), University Writing and Communication (Pass)

Highlights

Scalable Medical Imaging & Scientific Foundation Model

Expertise: MRI data processing, Large Vision-language Models (LVLM), Instruction Tuning, Reinforcement Learning from Human Feedback (RLHF), Reinforcement Learning on Chain-of-Thought (CoT) Reasoning, Contrastive Learning, Off-policy Knowledge Distillation, and On-policy Distillation (OPD).

- Vision-language Framework for Multi-sequence Brain Magnetic Resonance Imaging [Product] [Paper]
  Diala Lteif, **Shuyue Jia**, Subhrangshu Bit, Artem Kaliaev, Asim Z. Mian, ProfileJuan E. Small, Balamurugan Mangaleswaran, Bryan A. Plummer, Sarah A Bargal, Rhoda Au, Vijaya B. Kolachalama [Supplementary Material]
  To be submitted
  Dataset scale: >850K multi-sequence brain MRIs, >73K patient visits, >1M instruction tuning data
  Model scale: 35B MoE (3B activated) for faster real-world model inference
  Compute scale: 4 B200 GPUs (183GB memory), 2 H200 NVL GPUs (143GB memory), and 4 RTX PRO 6000 GPUs (97GB memory)
  Our vision: Clinically validated specialized intelligence for high-stakes medical imaging.

- Gradient-guided Adapter Merging for Neuroimaging Vision-language Models [Paper]
  Subhrangshu Bit, Osman B. Guney, Shuyue Jia, Vijaya B. Kolachalama *
  Technical Report, This work extends ReMIND toward personalized clinical intelligence.

- MedSyn: Text-guided Anatomy-aware Synthesis of High-fidelity 3D CT Images [Paper] [Codes]
  Yanwu Xu, Li Sun, Wei Peng, **Shuyue Jia**, Katelyn Morrison, Shyam Visweswaran, Motahhare Eslami, Kayhan Batmanghelich *
  *IEEE Transactions on Medical Imaging (IEEE T-MI, **IF in 2024: 8.9**)*

- PodGPT: An Audio-augmented Large Language Model for Research and Education [Product] [Paper]
  Shuyue Jia, Subhrangshu Bit, Edward Searls, Meagan Lauber, Lindsey Claus, Pengrui Fan, Varuna Jasodanand, Divya Veerapaneni, William Wang, Rhoda Au, Vijaya B. Kolachalama * [Supplementary Material]
  Source Codes: [**PodGPT Library**] [**PodRAG Pipeline**]
  Nature npj Biomedical Innovations
  Dataset scale: >42M text tokens, >3700 hours of podcasts, >10K top journal articles
  Model scale: 2B, 7B, 11B, 8×7B MoE, 70B
  Compute scale: >12 L40S GPUs (48GB memory)
  Our vision: Podcast-trained AI for equitable access to scientific knowledge.

- MedPodGPT: A Multilingual Audio-augmented Large Language Model for Medical Research and Education
  Shuyue Jia, Subhrangshu Bit, Edward Searls, Lindsey Claus, Pengrui Fan, Varuna Jasodanand, Meagan Lauber, Divya Veerapaneni, William Wang, Rhoda Au, Vijaya B. Kolachalama * [Product] [Paper] [Codes]
  Technical Report, This work was extended to PodGPT for STEMM.
  Dataset scale: >39M text tokens, >4300 hours of podcasts
  Model scale: 2B, 7B, 8B, 8×7B MoE, 70B
  Compute scale: >12 L40S GPUs (48GB memory)
  Our vision: Audio-augmented AI for medical research and education.

- Performance of Open-source Large Language Models on Nephrology Self-assessment Program
  Meysam Ahangaran, **Shuyue Jia**, Shlok Chitalia, Ambarish Athavale, Jean M. Francis, Michael William O'Donnell, Santhoshi Rupa Bavi, Uma Datta Gupta, Vijaya B. Kolachalama * [Paper] [Codes]
  Technical Report, This work extends the evaluation of PodGPT to Nephrology Self-Assessment Program (NephSAP).

AI-powered Autonomous Technology

Expertise: Agentic Framework, Agentic Retrieval-augmented Generation (RAG), Multi-agent Swarm (MAS), Tool Using, and Automatic AI Workflow, Generative Models, Large Language Models (LLMs), Continual Pre-training, and Retrieval-augmented Generation (Retrieval, Reranking, and Generation).

- Agentic Memory-augmented Retrieval and Evidence Grounding for Medical Question-answering Tasks [Paper]
  Shuyue Jia, Subhrangshu Bit, Varuna Jasodanand, Yi Liu, Vijaya B. Kolachalama * [Supplementary Material]
  *International Journal of Medical Informatics (IJMI, **IF in 2026: 5.0**)*
  Dataset scale: >30M documents, >68M snippets
  Model scale: 72B (equipped with 5 designed tools)

Compute scale: 4 L40S GPUs (48GB memory)
Our vision: AI-powered precision medical evidence for everyone, everywhere.

- DocAgent: An Agentic Framework for Multi-modal Long-context Document Understanding [Paper] [**Codes**]
  Li Sun, Liu He, **Shuyue Jia**, Yangfan He, Chenyu You *
  The 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP 2025 Main Conference)

Computer Vision

Expertise: Image Quality Assessment (IQA) and general Image and Video Processing, including but not limited to image enhancement (super-resolution, denoising, deblurring), and image segmentation.

- No-reference Image Quality Assessment via Non-local Dependency Modeling [Paper] [Codes] [Slides] [Poster]
  Shuyue Jia, Baoliang Chen, Dingquan Li, Shiqi Wang *
  IEEE 24th International Workshop on Multimedia Signal Processing (IEEE MMSP'22) (**Poster Presentation**)

- Learning from Mixed Datasets: A Monotonic Image Quality Assessment Model [Paper] [Codes]
  Zhaopeng Feng, Keyang Zhang, **Shuyue Jia**, Baoliang Chen, Shiqi Wang *
  IET Electronics Letters
  Awarded Wiley Top Cited Article 2023-2024

Deep Learning for Neural Systems and Brain Dynamics

Expertise: EEG Signal Processing and general time-series signal processing.

- GCNs-Net: A Graph Convolutional Neural Network Approach for Decoding Time-resolved EEG Motor Imagery Signals [Paper] [Codes] [Slides] [Survey]
  Shuyue Jia, Yimin Hou, Xiangmin Lun, Ziqian Hao, Yan Shi, Yang Li, Rui Zeng, Jinglei Lv *
  IEEE Transactions on Neural Networks and Learning Systems (IEEE T-NNLS)
  IF in 2022: 14.255, Top 0.1% most cited articles published in Engineering in 2022
  GitHub Repo obtained **1K+ stars and 200+ forks**

- Deep Feature Mining via Attention-based BiLSTM-GCN for Human Motor Imagery Recognition
  Shuyue Jia *, Yimin Hou, Xiangmin Lun, Shu Zhang, Tao Chen, Fang Wang, Jinglei Lv [Paper] [Codes] [Slides]
  Frontiers in Bioengineering and Biotechnology
  IF in 2021: 6.064, Top 1% most cited articles published in Engineering in 2022

- A Novel Approach of Decoding EEG Four-class Motor Imagery Tasks via Scout ESI and CNN [Paper] [Codes]
  Yimin Hou, Lu Zhou *, **Shuyue Jia**, Xiangmin Lun
  Journal of Neural Engineering
  IF in 2020: 5.379, Top 1% most cited articles published in Engineering in 2020
  GitHub Repo obtained **200+ stars and 40+ forks**

Intelligent Technology

- PMU Measurements based Short-term Voltage Stability Assessment of Power Systems via Deep Transfer Learning [Paper]

Yang Li *, Shitu Zhang, Yuanzheng Li, Jiting Cao, **Shuyue Jia**
IEEE Transactions on Instrumentation and Measurement (IEEE T-IM)
IF in 2023: 5.6, Top 1% most cited articles published in Engineering in 2023

- Improving Performance: A Collaborative Strategy for the Multi-data Fusion of Electronic Nose and Hyperspectral to Track the Quality Difference of Rice [Paper]
Yan Shi, Hangcheng Yuan, Chenao Xiong, **Shuyue Jia**, Jingjing Liu, Hong Men *
Sensors & Actuators: B. Chemical
IF in 2021: 9.221, Top 10% most cited articles published in Engineering

Thesis

- No-reference Image Quality Assessment via Non-local Modeling [Permanent Link] [Defense Slides]
Shuyue Jia
M.Phil. Thesis, City University of Hong Kong, May 2023.

- Brain-computer Interface Signals Classification and Its Applications based on Deep Learning Methods
Outstanding Thesis Award [Related Publication] [Defense Slides]
Shuyue Jia
B.Eng. Thesis, Northeast Electric Power University, May 2020.

Note:
1. Google Scholar Profile
2. * denotes the Corresponding Author

Selected Projects

Vision-language Framework for Multi-sequence Brain MRIs — 2025 - 2026

Mentor: Prof. Vijaya B. Kolachalama, Boston University

- **Dataset scale**: >850K multi-sequence brain MRIs, >73K patient visits, >1M instruction tuning data
- **Model scale**: 35B MoE (3B activated), 27B, and 9B
- **Neuroimaging (MRI) data processing & harmonization**: DICOM → NIfTI → 2D/3D/4D volumes → MONAI RAS anatomical orientation → Move through-plane axis (largest voxel spacing) → Intensity normalization → Square padding & resizing → Slices to frames with discarding all-black slices, uint8 scaling, and radiological orientation → Frames to lossless video
- **Instruction tuning dataset curation**: Introduced 12 question types (as tasks), including 6 classification-based QA types (Choice Verification, Yes/No QA, True/False, Multiple Choice QA, Minimal Pair Discrimination, and Textual Entailment, with answer distribution balanced) and 6 generation-based QA types (Classification, Extraction, Cloze (Masked Word), Factual QA, Fill-in-the-Blank, and Sentence Completion).
- **Model training**: 35B MoE (3B activated): Hugging Face trainer with Flash Attention 2, causal-conv1d kernel, and torch compiled on 4 B200 GPUs (183GB memory, Blackwell, CUDA 12.8), 27B Dense model: Flash Attention 3, causal-conv1d kernel, and torch compiled on 2 H200 NVL GPUs (143GB memory, Hopper, CUDA 12.8), and 9B Dense model: Flash Attention 2, causal-conv1d kernel, and torch compiled on 2 RTX PRO 6000 GPUs (97GB memory, Blackwell, CUDA 13.2)
- **Model evaluation & inference**: vLLM generate engine (0.17.0+cu132) compiled on 2 RTX PRO 6000 GPUs (97GB memory, CUDA 13.2).

- **Model deployment** (served on https://remind.bu.edu): 35B MoE (3B activated) with vLLM serve engine (0.24.0) on 4 RTX 3090 GPUs (24GB memory, CUDA 13.2).

Continual Pre-training of LLMs via Podcast Audio, RAG, & Agentic RAG 2024 - 2025

Mentor: Prof. Vijaya B. Kolachalama, Boston University

- **Dataset scale**: >42M text tokens, >3700 hours of podcasts, >10K top journal articles
- **Model scale**: 2B, 7B, 11B, 8×7B MoE, 70B
  Model training & inference: 12 L40S GPUs (48GB memory, CUDA 12.8) with vLLM generate engine (0.6.2)
- **Computational framework**: Developed a computational framework designed to improve Large Language Models (LLMs) leveraging the informational content of publicly accessible STEMM podcasts. [Product] [Paper] [Codes]
- **Dataset curation**: Collected and processed Creative Commons Attribution (CC BY) licensed and NEJM podcasts from scratch.
- **Model training & evaluation & inference**: Created and maintained **GitHub Library PodGPT**, a whole pipeline that consists of Continual Pre-training, Low-Rank Adaptation (LoRA) for Efficient Fine-Tuning, GPTQ Model Quantization, STEMM Benchmarking, and real-world deployment by vLLM serve engine.
- **RAG pipeline for AI grounding**: Introduced and maintained **PodRAG**, a **Retrieval-Augmented Generation (RAG)** pipeline featuring retrieval and reranking using PubMed Central and *The New England Journal of Medicine* (NEJM) articles, designed to reduce hallucinations and provide grounding through fact-checking.
- **April 2025 update**: We extended this work to a unified, open-source 72B LLM-based agentic system (served on 4 L40S GPUs, vLLM 0.6.3 serve, openai[aiohttp]-compatible server), integrating document retrieval, re-ranking, evidence grounding, and diagnosis generation to support dynamic, multi-step medical reasoning.
  Updated dataset scale: >30M documents, >68M snippets [Paper]
- **April 2026 update**: We extended PodGPT's evaluation on Nephrology Self-Assessment Program (NephSAP) multiple-choice QA dataset with clinical validations by 3 board-certified nephrologists (5 questions ×8 nephrology subject areas) to better understand its capabilities and limitations within this specialized clinical domain. [Paper] [Inference Code] [Figures Code]

Non-local Modeling for Image Quality Assessment 2021 - 2023

Mentor: Prof. Shiqi Wang, City University of Hong Kong

- Built a **Superpixel-based Graph Neural Network** model to explore the non-local interactions in quality prediction. [Paper] [Codes]
- Presented the definitions of **Global Distortions** and **Local Distortions** for IQA. [Thesis] [Slides]

Graph Representation Learning for EEG-based BCI Classification 2019 - 2020

Mentor: Prof. Yimin Hou, Northeast Electric Power University and Prof. Jinglei Lv, The University of Sydney

- Implemented a **Convolutional Neural Network** architecture for Morlet wavelet joint time-frequency analysis of EEG time series. [Paper] [Codes] (**obtained 200+ stars and 40+ forks**)
- Introduced **Graph Modeling** and **Graph Representation Learning** to decode raw EEG signals by cooperating with the functional topological relationship of electrodes. [Paper] [Codes]
- Proposed a remarkably accurate and responsive recognition framework via **Bidirectional Long short-term Memory (BiLSTM)** with the **Attention Mechanism** and **Graph Neural Networks**. [Paper] [Codes]
- Created and maintained GitHub Library **EEG-DL** (**obtained 1K+ stars and 200+ forks**), a Deep Learning library written by TensorFlow for EEG tasks classification.

Statistical Measures for Sonar Image Segmentation 2018 - 2019

Mentor: Prof. Shengxi Jiao, Northeast Electric Power University

- Engineered a framework to enhance the contract between the ship and the reverberation area, which consists of **Image Filtering** by discrete cosine transform (DCT) → **Edge Detection** by the Roberts cross operator → **Object Localization** by thresholding 1D histogram → **Morphological Local Enhancement** by an expansion operator.
- Significantly improved the segmentation accuracy (86%) compared with that without the pre-processed stage (80%).
- A patent is issued based on the project and Codes.

Industry Experience

Philips Research, Shanghai, China **Natural Language Processing Intern** Jul - Oct 2020

Mentor: Dr. Shuang Zhou, Precision Diagnosis & Image Guided Therapy (PD&IGT) Department.

- **Medical Concept Mapping**: three levels to map a query to the standard term → **Byte-pair Encoding** and **FMM & BMM Algorithms** for sub-words generation and matching (syntax-level), **Word2vec Cosine Similarity** (semantics-level), and **Knowledge Graph** (pragmatics-level).
- **Medical Named Entity Recognition (NER)**: compared the performances of different models → CRF++, Character-level BiLSTM + CRF, Character-level BiLSTM + Word-level BiLSTM / CNNs + CRF, and deployed the models using Python Flask and Docker as a web app.
- **Dynamic Webs Crawling**: crawled >620K pages from the China National Science and Technology Library (NSTL) via Python threading.

Teaching Appointments

EC327: Introduction to Software Engineering, Boston University, MA, USA Spring 2025

- Introduction: This course aims to introduce students to software design, object-oriented programming techniques, elementary data structures and algorithmic analysis, and software engineering principles.
- Responsibility: Prepared lab session materials, led lab sessions for students, and designed homework assignments with automated grading.

Software, Library, and Toolkit

Software - **ReMIND** [Product] [Paper]

- An online platform for deploying our latest video foundation model for multi-sequence brain MRIs.

Software - **PodGPT** [Product] [Paper] [Codes]

- An online platform for deploying our latest multimodal foundation models for medical and clinical applications.
- The success of the PodGPT project is a result of **collaborative teamwork** by a group of members of the Kolachalama Laboratory. The PodGPT Library is maintained by the Kolachalama Laboratory at Boston University.

Library - **PodGPT Library** [Paper] [Codes]

- A library for benchmarking multilingual medical Large Language Models (LLMs).

Library - **EEG Deep Learning Library** (**1K+ stars and 200+ forks**) [Paper] [Codes]

- A Deep Learning (DL) library written by TensorFlow for EEG tasks (signals) classification.

Toolkit - **PromptCraft** [Codes] [PyPI Package]

- A prompt perturbation toolkit from the character, word, and sentence levels for prompt robustness analysis.

- Usage: `pip install promptcraft`

Invited Reviewer

Conducted over 150 technical and multidisciplinary paper reviews for top-tier journals and computer science conferences, contributing to the advancement of research and maintaining high publication standards.

IEEE Transactions on Pattern Analysis and Machine Intelligence (T-PAMI)
IEEE Transactions on Multimedia (T-MM)
IEEE Transactions on Circuits and Systems for Video Technology (T-CSVT)
IEEE Transactions on Neural Networks and Learning Systems (T-NNLS)
IEEE Transactions on Medical Imaging (T-MI)
IEEE Transactions on Industrial Informatics (T-II)
IEEE Transactions on Biomedical Engineering (T-BME)
IEEE Transactions on Human-Machine Systems (T-HMS)
IEEE Transactions on Affective Computing (T-AFFC)
IEEE Transactions on Cognitive and Developmental Systems (T-CDS)
IEEE Transactions on Emerging Topics in Computational Intelligence (T-ETCI)
IEEE Journal of Biomedical and Health Informatics (J-BHI)
IEEE Journal of Selected Topics in Signal Processing (J-STSP)
IEEE Open Journal of the Industrial Electronics Society (OJ-IES)
IEEE Open Journal of the Computer Society (OJ-CS)
IEEE MultiMedia (MM)
IEEE Sensors Journal
Computer Vision and Image Understanding (CVIU)
Journal of Medical Internet Research (JMIR)
IET Image Processing
IET Science, Measurement & Technology
Neural Information Processing Systems (NeurIPS) 2026
Association for Computational Linguistics (ACL) Industry Track 2026
International Conference on Learning Representations (ICLR) 2025, 2026

Conference Planning and Preparation (Steering Committee Member)

2027 IEEE International Conference on AI and Data Analytics (IEEE ICAD) June 2027
Venue: TBD
2026 IEEE International Conference on AI and Data Analytics (IEEE ICAD) June 2026
Venue: Fulton Hall & Stokes Hall, Boston College, 140 Commonwealth Ave., Chestnut Hill, MA, USA
2025 IEEE International Conference on AI and Data Analytics (IEEE ICAD) June 2025
Venue: Joyce Cummings Center, Tufts University, 177 College Ave., Medford, MA, USA

Professional Skills

Coding: Vibe Coding, Python, Matlab, R, C/C++, HTML, CSS
Library: TensorFlow, PyTorch, Scikit-learn, Hugging Face (Transformers, PEFT, Datasets, etc.), vLLM
Tools: Git, SVN, Unix Shell, Vim, Markdown, LaTeX, Docker, K8s
Hardware: Nvidia Jetson Nano (Ubuntu), Raspberry Pi (Raspbian OS), 80C52 Microcontroller (Embedded C)
Blogs: Personal Writings and Blogs

Photography: Bruce S. JIA Photography, Boston - Hong Kong - Beijing